

Frequencies and Rhythms

As the initial step toward interpretation, the reader should clearly understand these two terms.

Frequency

Frequency is the number of times a waveform occurs per second (one big square). If a waveform occurs three times in a second, then its frequency is 3 Hz. If it occurs once over 2 s, then its frequency is 0.5 Hz. Therefore, the frequency simply describes the rate at which a waveform occurs, it doesn't indicate if that waveform is normal or abnormal. Conventionally, electroencephalographers group frequencies into standardized ranges (or bands) based on the Greek numeral.

They include delta, theta, alpha, beta, and gamma in increasing order of frequency (these exact cut offs may vary):

- Delta activity occurs with a frequency of 0.5 to 4 Hz.
- Theta activity occurs with a frequency of 5 to 7 Hz.
- Alpha activity occurs with a frequency of 8 to 13 Hz.
- Beta activity occurs with a frequency of 14 to 30 Hz.

Most electrographic activity on scalp EEG occurs between 0.5 to 30 Hz. The amplitude of the activity is usually inversely related to its frequency so lower frequencies typically have higher amplitudes compared to faster frequencies, but this isn't a rule.

Electrographic activities with frequencies above 30 Hz are called high-frequency oscillations (HFOs). These include gamma activity (30–80 Hz), ripples (80–200 Hz) and fast ripples (200–500 Hz). HFOs are difficult to appreciate on scalp EEG. Frequencies below 0.5 Hz are called infra-slow oscillations (IFOs) and are seen in preterm neonates [1].

Rhythm

A rhythm is a specific pattern of electrographic activity with a characteristic frequency. Therefore, in addition to frequency, rhythms have a typical location, morphology, and reactivity. Rhythms are also dependent on the

physiological state such as wakefulness, drowsiness, or sleep (state dependence). Unlike frequency, rhythms indicate if the EEG is normal or abnormal. The reader should recognize the common rhythms and understand their implications [1].

As an example, alpha activity is not the same as an alpha rhythm. Any electrographic activity that occurs within the range of 8 to 13 Hz (alpha range) is termed *alpha activity*. However, the *alpha rhythm* specifically refers to the 8–13 Hz (alpha frequency) pattern with a posterior dominance (location) and sinusoidal appearance (morphology) that attenuates with eye opening (reactivity) that is seen during normal wakefulness (state dependence). Furthermore, alpha activity may be normal or abnormal, but alpha rhythm is the electrographic hallmark of normal wakefulness.

Alpha Rhythm

Recognizing the alpha rhythm is the starting point of all EEG interpretation. Historically, this was the first rhythm to be identified and it is thought to represent the resting activity of the occipital cortex. As mentioned in the previous example; during normal wakefulness, a posterior-predominant pattern of 8 to 13 Hz frequency (alpha frequency) occurs symmetrically over the occipital regions – this is the “normal” posterior-dominant rhythm (PDR). When this posterior-dominant rhythm attenuates with eye opening (reactivity) it is called the “alpha rhythm” – an obligate feature of normal adult wakefulness.

The posterior-dominant rhythm first attains a frequency of about 8 Hz at approximately 2–3 years of age will remain stable within the alpha range (commonly 9–10 Hz) till about the ninth decade of life, after which there may be a slight decline. The alpha rhythm is best appreciated in a relaxed state with eyes closed and attenuates with eye opening or intellectual effort. Figure 6.1 shows an 11 Hz alpha rhythm.

Normal Variations of the Alpha Rhythm

Alpha Squeak

This is a transient increase in the frequency of the alpha rhythm immediately after eye closure. Predigital era EEG machines had ink pens, these produced a squeaking noise each time the patient blinked, hence its name.

The reader must count the frequency of the posterior-dominant rhythm a second after eye closure to avoid overestimating it (avoid the squeak). Figure 6.1 shows the correct method to count the frequency of the posterior-dominant rhythm while Figure 6.2 shows prominent alpha squeak.

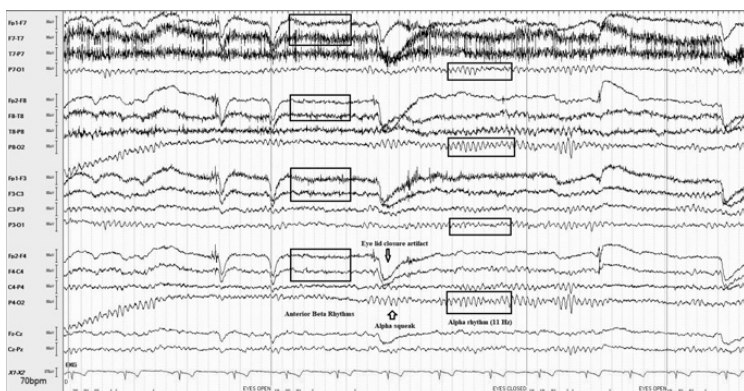


Figure 6.1 Twenty-seven-year-old woman with anxiety, anterior beta, and posterior alpha rhythms.

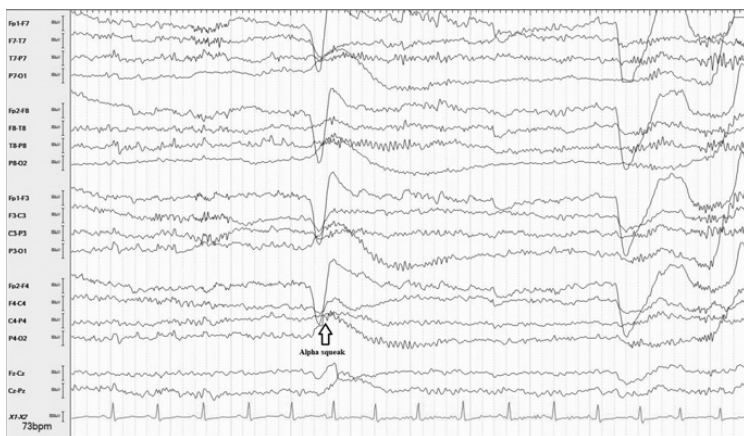


Figure 6.2 Thirty-four-year-old woman with staring spells, prominent alpha squeak.

Amplitude

The typical amplitude (voltage) of the alpha rhythm is 30–40 μV , but there is considerable variation. Lower amplitudes may result in poor visualization of

the alpha rhythm in as many as a quarter of normal adults. In these circumstances, the sensitivity may be increased.

Symmetry

A slight asymmetry of alpha amplitude and frequency between the two hemispheres is normal. The dominant hemisphere (commonly left) may have a slightly lower amplitude compared to the nondominant side (commonly right).

Paradoxical Alpha

Normally, the alpha rhythm attenuates with eye opening. Sometimes during drowsiness, it may paradoxically increase with eye opening – paradoxical alpha.

Slow and Fast Alpha Variants

Alpha variants include slow and fast variants that show a harmonic relationship but with similar distribution and reactivity. Slow variants have frequencies typically half that of alpha (4–5 Hz) and fast variants have frequencies twice that of alpha (16–20 Hz). Occasionally, these variants may have a notched appearance [2]. Figure 6.3 shows a slow alpha variant.

Abnormalities of the Alpha Rhythm

The alpha rhythm may be abnormal if it is slow, asymmetric or unreactive (Bancaud's phenomena).

Generalized Slowing: This occurs if the alpha rhythm does not exceed 8 Hz at any point during the recording. Though 8 Hz is the lower normal of alpha range, this frequency is thought to occur in less than 1% of normal adults. Figure 6.4 shows generalized slowing.

Asymmetry: Amplitude asymmetry of greater than 50% is abnormal especially if the dominant hemisphere (commonly left) is greater than the nondominant hemisphere (commonly right). Normally, the frequency between hemispheres differs by less than 0.5–1 Hz. Asymmetries of more than 1 Hz are abnormal.

Bancaud's Phenomenon: This refers to the failure of the alpha rhythm to attenuate with eye opening on the side ipsilateral to a structural abnormality. This is usually associated with occipital or pontine lesions [2].

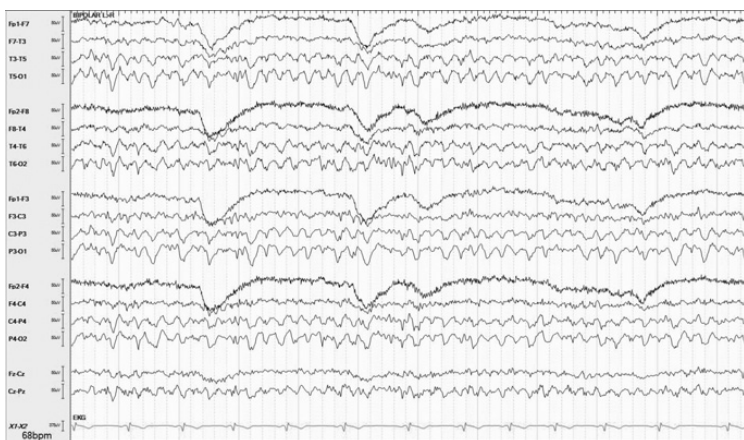


Figure 6.3 Twenty-seven-year-old man with syncope, slow alpha variant (4 Hz).

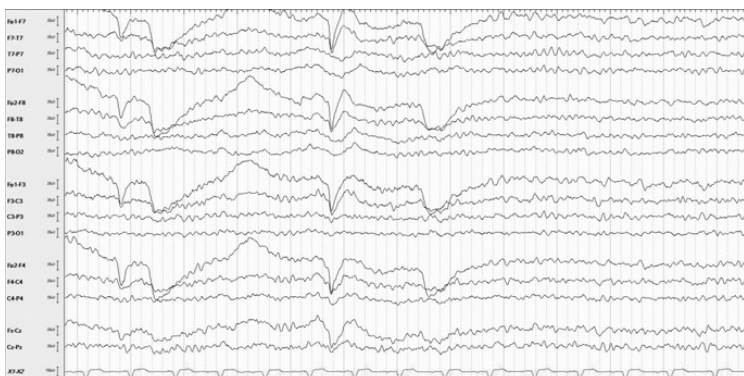


Figure 6.4 Seventy-two-year-old woman with dementia, mild generalized alpha slowing (8 Hz).

Beta Rhythm

Beta rhythms have frequencies of 14 Hz or greater (commonly 18–25 Hz). They have a low amplitude (less than 20 μ V) and an anterior predominance.

Beta rhythms are thought to represent the resting activity of the precentral cortex; they tend to attenuate with contralateral movements (much like the “mu” rhythm which is described later in Chapter 13). Normally, beta rhythms occur during wakefulness and accentuate during drowsiness and light sleep. Speech and intellectual efforts also accentuate beta rhythms. Figure 6.1 shows the normal anterior beta rhythm during wakefulness.

Excessive Beta

This is usually the result of sedatives such as benzodiazepines and barbiturates. There may be increased amplitude ($>25 \mu\text{V}$), prevalence ($>50\%$ of the record) and/or generalized distribution of beta rhythm. Figure 6.5 shows excessive beta associated with benzodiazepine use.

Asymmetric Beta

The persistent suppression of beta over a region or hemisphere may indicate an ipsilateral cortical abnormality or an interruption between the cortical surface and the recording electrode such as a subdural hematoma or scalp edema. Conversely, breach effect may result in an increase in beta amplitudes due to loss of skull bone [2].

Theta Rhythm

Theta rhythms have frequencies of 5 to 7 Hz with varying amplitudes and morphology. A third of teens and young adults normally have 6 to 7 Hz theta over the frontocentral region during wakefulness. This may be enhanced with

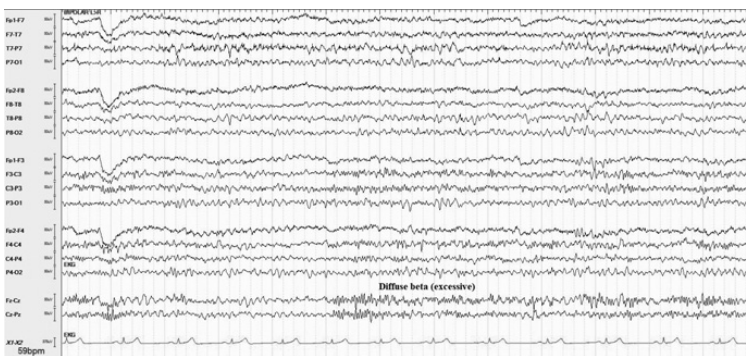


Figure 6.5 Fifty-six-year-old woman with epilepsy on clobazam, diffuse beta rhythms (excessive beta).

emotions, focused concentration, or mental tasks. Also, a third of asymptomatic elderly will have intermittent 5–6 Hz theta over the bitemporal regions (usually left greater than right).

However, in most normal adults, theta rhythms are limited to drowsiness or light sleep and their presence during wakefulness is abnormal [2]. Midline theta may occur as a normal variant as described in Chapter 13.

Delta Rhythm

Delta rhythms have frequencies below 4 Hz. In young children, their prevalence normally doesn't exceed 10% of the waking record and in the elderly, rare intrusions of isolated delta (often over the left temporal region) during drowsiness may be permissible.

However, in most normal adults, delta rhythms are limited to sleep and their presence during wakefulness is abnormal. Focal delta may indicate underlying cerebral lesions and diffuse delta may indicate generalized cerebral dysfunction or encephalopathy [2]. Figure 6.6 shows theta and delta rhythms in encephalopathy.

Chapter Summary

1. Frequency is the number of times a waveform occurs per second (one big square).
2. Rhythms in addition to frequency have characteristic features of location, morphology, reactivity, and state dependence.
3. Frequencies are descriptive; rhythms are diagnostic.

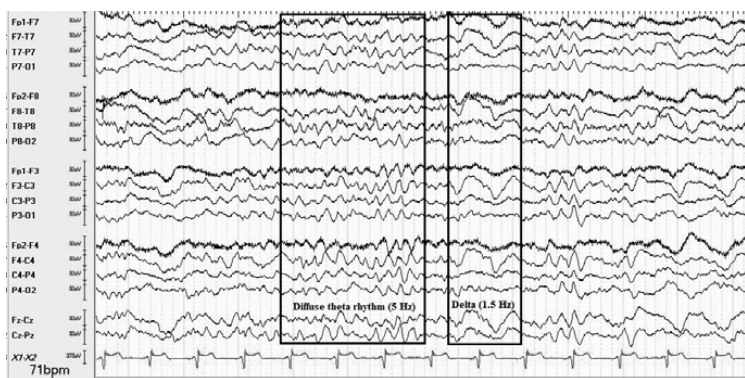


Figure 6.6 Sixty-two-year-old man with hepatic failure, diffuse theta and delta rhythms.

4. Clinically relevant frequency bands include delta (0.5–4 Hz), theta (5–7 Hz), alpha (8–13 Hz), and beta (14–30 Hz).
5. The alpha rhythm is an obligate feature of normal wakefulness; it has considerable variations.
6. Abnormal alpha rhythm may be slow, asymmetric, or unreactive (Bancaud's phenomena).
7. Excessive beta is associated with sedative medications.
8. In most normal adults, theta and delta rhythms are limited to drowsiness or sleep; their occurrence during wakefulness may be abnormal.

References

1. Nayak CS, Anilkumar AC. EEG normal waveforms. *InStatPearls* [Internet] 2020 Jun 28.
2. Tatum WO IV, Husain AM, Benbadis SR, Kaplan PW. Normal adult EEG and patterns of uncertain significance. *Journal of Clinical Neurophysiology*. 2006 Jun 1;23(3):194–207.